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روزبه پورجعفریان (۴۰۴۲۰۱۹۸۴)

ریاضی مهندسی پیشرفته

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تاریخ دریافت: ۱۴۰۴/۰۹/۲۷

تاریخ ارسال: ۱۴۰۴/۱۰/۰۹

تمرین شمارهٔ ۳

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1 $X \in \mathbb{R}^{N \times P}$, $x_1, \dots, x_n \in \mathbb{R}^P$, φ: single unit vector $\in \mathbb{R}^P$, $z_i = x_i^T \varphi$ $i = 1, \dots, N$

$\hat{x}_i = z_i \varphi \Rightarrow$ error: $L(\varphi) = \sum_{i=1}^{N} \|x_i - \hat{x}_i\|^2$

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a) show: $L(\varphi) = \sum_{i=1}^{N} \|x_i\|^2 - \sum_{i=1}^{N} (x_i^T \varphi)^2$

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by: $\|a - b\|^2 = (a-b)^T (a-b) = a^T a - 2a^T b + b^T b$

for point i can write:

$$\|x_i - \hat{x}_i\| = \|x_i - z_i \varphi\| = \|x_i - (x_i^T \varphi)\varphi\| = \overset{①}{x_i^T x_i} - \overset{②}{2x_i^T ((x_i^T \varphi)\varphi)} + \overset{③}{((x_i^T \varphi)\varphi)^T ((x_i^T \varphi)\varphi)}$$

①: $x_i^T x_i = \|x_i\|^2$

②: $-2x_i^T ((x_i^T \varphi)\varphi) = -2((x_i^T \varphi)(x_i^T \varphi)) = -2(x_i^T \varphi)^2$

③: $((x_i^T \varphi)\varphi)^T ((x_i^T \varphi)\varphi) = \varphi^T (x_i^T \varphi)^T (x_i^T \varphi)\varphi = (x_i^T \varphi)^2 \|\varphi\|^2 = (x_i^T \varphi)^2$

$\Rightarrow \|x_i - \hat{x}_i\| = \|x_i\|^2 - 2(x_i^T \varphi)^2 + (x_i^T \varphi)^2 = \|x_i^2\| - (x_i^T \varphi)^2$

$\Rightarrow \sum_{i=1}^{N} \|x_i - \hat{x}\|^2 = \sum_{i=1}^{N} \left(\|x_i\|^2 - (x_i^T \varphi)^2\right) = \sum_{i=1}^{N} \|x_i\|^2 - \sum_{i=1}^{N} (x_i^T \varphi)^2$

$\Rightarrow L(\varphi) = \sum \|x_i\|^2 - \sum (x_i^T \varphi)^2$

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b) min error $\Rightarrow$ max $\sum_{i=1}^{N} (x_i^T \varphi)^2$

$$L(\varphi) = \underbrace{\sum_{i=1}^{N} \|x_i\|^2}_{\mathrm{I}} - \underbrace{\sum_{i=1}^{N} (x_i^T \varphi)^2}_{\mathrm{II}}$$

$$\left.\begin{array}{l} \text{(I) depends only on } x_i \rightarrow \text{constant} \rightarrow \sum \|x_i\| = C \\ \text{(II) depends on } \varphi \rightarrow S(\varphi) = \sum (x_i^T \varphi)^2 \end{array}\right\} \Rightarrow L(\varphi) = C - S(\varphi)$$

if $S(\varphi)$ increase, $L(\varphi)$ decrease and if $S(\varphi)$ decrease, $L(\varphi)$ increase.

$$\arg\min_{\|\varphi\|=1} L(\varphi) = \arg\min_{\|\varphi\|=1} C - S(\varphi) = \arg\max_{\|\varphi\|=1} S(\varphi) = \arg\max_{\|\varphi\|=1} \sum (x_i^T \varphi)^2$$

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c) Solution φ is eigenvector of covariance matrix to longest eigenvector

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$$\sum_{i=1}^{N} (x_i^T \varphi)^2 = \sum_{i=1}^{N} (x_i^T \varphi)^T (x_i^T \varphi) = \sum_{i=1}^{N} \varphi^T x_i x_i^T \varphi = \varphi^T \underbrace{\left(\sum x_i x_i^T\right)}_{NS} \varphi \quad , S = \frac{1}{N} \sum x_i x_i^T$$

$\sum (x_i^T\varphi)^2 = N\varphi^T S\varphi \xrightarrow{N=cte} \arg\max_{\|\varphi\|=1} \sum (x_i^T\varphi)^2 = \arg\max_{\|\varphi\|=1} \varphi^T S\varphi$

like $\max b^T S b$ s.t $\|b\|=1$

$\left.\begin{matrix}\max \varphi^T S\varphi \\ \|\varphi\|=1\end{matrix}\right\} \rightarrow L(\varphi,\lambda) = \varphi^T S\varphi + \lambda(1-\varphi^T\varphi)$

$\nabla_\varphi L = 2S\varphi - 2\lambda\varphi = 0 \rightarrow S\varphi = \lambda\varphi$

$\rightarrow \varphi^T S\varphi = \varphi^T(\lambda\varphi) = \lambda(\varphi^T\varphi) = \lambda \rightarrow$ to max $\varphi^T S\varphi$, should chose eigenvector as large as possible and eigen vector corresponding to largest eigenvalue.

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2 $\min_{\beta,\beta_0,\xi} \frac{1}{2}\|\beta\|^2 + C\sum_{i=1}^{N}\xi$

s.t $y_i(\beta_0 + x_i^T\beta) \geq 1-\xi_i \quad i=1,\dots,N$

$\xi_i \geq 0$ Soft Margin SVM

Binary classification dataset

$(x_i,y_i) \in \mathbb{R}^p \; \{-1,+1\} \quad i=1,\dots,N$

$M = 1/\|\beta\|$

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a) if $0<\xi_i<1$, sample (x_i,y_i) lies inside the margin and correctly classified

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$\begin{cases} f(x) = \beta_0 + x^T\beta \\ t_i = y_i f(x_i) = y_i(\beta_0 + x_i^T\beta) \\ t_i \geq 1-\xi_i, \quad \xi_i \geq 0 \end{cases}$

$0<\xi_i<1 \rightarrow -1<-\xi_i<0 \rightarrow 0<1-\xi_i<1$, $t_i \geq 1-\xi_i \rightarrow t_i>0$

if $t_i>0 \rightarrow y_i f(x_i)>0 \rightarrow$ same sign and base on $y_i \in \{-1,1\} \rightarrow$ correct classify ✓

for margin:

$t>1$: correct side, out of margin

$0<t<1$: correct side, inside margin $\Rightarrow t_i<1$ inside the margin

$t<0$: wrong side

$\xi_i = \max\{0, 1-t_i\} \rightarrow$ the slack ξ_i show how much to margin constraint is violated.

minimizing hing loss is equivalent to introducing $\xi \geq 0$ with $\xi \geq 1-t$

if $0 < \xi_i < 1$ then $\xi_i > 0 \Rightarrow \xi_i = 1 - t_i \Rightarrow t_i = 1 - \xi_i$

and $0 < \xi_i < 1 \Rightarrow 0 < 1 - \xi_i < 1 \Rightarrow 0 < t_i < 1$

$\Rightarrow$ $0 < t_i = y_i(\beta_0 + x_i^T\beta) < 1$, $t_i > 0$: correctly classified

$t_i < 1$: inside the margin

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b) if $\xi_i > 1 \Rightarrow (x_i, y_i)$ is missclassified

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if $\xi_i > 1 \Rightarrow 1 - \xi_i < 0 \Rightarrow t_i \geq 1 - \xi_i$ → negative value

and, $t_i \geq 1 - \xi_i \Rightarrow \xi_i \geq 1 - t$, $\xi_i \geq 0 \Rightarrow \xi_i = \max\{0, 1 - t_i\}$

if $\xi_i > 1 \Rightarrow \xi_i = 1 - t_i \Rightarrow \xi_i > 1 \Rightarrow 1 - t_i > 1 \Rightarrow t_i < 0$

$\Rightarrow t_i = y_i f(x_i) < 0$ different signs $\Rightarrow$ misclassified

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c) if every point satisfy $\xi_i = 0 \Rightarrow M = \frac{1}{\|\beta\|}$, and if $\xi_i > 0 \Rightarrow M \leq 1/\|\beta\|$

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if $\xi_i = 0 \Rightarrow y_i f(x_i) \geq 1$ $i = 1, \dots, N$

distance of a point to the decision boundary :

D.B: $\beta_0 + x^T\beta = 0 \Rightarrow \text{dist}(x_i, \{\beta_0 + x^T\beta = 0\}) = \frac{|\beta_0 + x_i^T\beta|}{\|\beta\|}$

multiply by $y_i = \pm 1$, $|a| = |y_i a| \Rightarrow \text{dis} = \frac{|y_i(\beta_0 + x_i^T\beta)|}{\|\beta\|} = \frac{t_i}{\|\beta\|} > 0$ ✓ (hard margin : $t_i > 0$)

Geometry margin is the closest any training point gets to the boundary: $M = \min \frac{t_i}{\|\beta\|}$

but $t_i \geq 1 \Rightarrow M \geq \frac{1}{\|\beta\|}$

if every point had $t_i > 1$, we could scale (β, β_0) down slightly and still keep $t_i \geq 1$ while reducing $\|\beta\|$. That could improve the objective. so at the optimum, at least one point must satisfy $t_i = 1$ (a support vector) $\Rightarrow$ $M = \min_i \frac{t_i}{\|\beta\|} = \frac{1}{\|\beta\|}$

if $\xi \geq 0 \Rightarrow t_i = y f(x_i) \geq 1 - \xi_i$, $\xi_i > 0 \Rightarrow 1 - \xi_i < 1 \Rightarrow t_i \leq 1$

in optimal solution with $\xi_i > 0$, we typically have equality $t_i = 1 - \xi_i \Rightarrow t_i < 1$

Distance of that point to the boundary: $\text{dis}_i = \frac{f(x_i)}{\|\beta\|} = \frac{|t_i|}{\|\beta\|}$

if $0 < \xi_i < 1 \Rightarrow 0 < t_i < 1 \Rightarrow \text{dist}_i = \frac{t_i}{\|\beta\|} < \frac{1}{\|\beta\|}$

if $\xi_i > 1$ then $t_i < 0$, still $\frac{t_i}{\|\beta\|}$ is distance. importantly the point is not achieving the margin band $t = 1$.

Either way, there exist a point whose signed margin value is below 1, so the min distance to the boundary cannot exceed $1/\|\beta\|$

So, $M = \min \frac{t_i}{\|\beta\|} \leq \frac{1}{\|\beta\|} \Rightarrow M \leq \frac{1}{\|\beta\|}$

In hard margin SVM, $\xi_i = 0$, every point must sit outside the margin band, $t_i \geq 1$. The closest point touch $t_i = 1$, so the margin become exactly $\frac{1}{\|\beta\|}$.

In soft margin, SVM, $\xi_i \geq 0$, we allow some points to violate the margin constraint, meaning some points can have $t_i < 1$ or even $t_i < 0$, once any point is allowed to move inside, the min distance to the boundary can only get smaller so the margin is at most $\frac{1}{\|\beta\|}$.

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3 $\min \frac{1}{2}\|\omega^2\|$

s.t. $y_i(\omega^T x_i + b) \geq 1 \quad \forall i = 1, \dots, n$

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a) Lagrangian $L(\omega, b, \alpha)$, $\alpha_i \geq 0$ Lagrangian multipliers

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$$\left.\begin{array}{l} \min \frac{1}{2}\|\omega\|^2 \\ 1 - y_i(\omega^T x_i + b) \leq 0 \end{array}\right\} \Rightarrow L(\omega, b, \alpha) = \frac{1}{2}\|\omega\|^2 + \sum_{i=1}^{n} \alpha_i\left(1 - y_i(\omega^T x_i + b)\right) \quad \alpha_i \geq 0$$

$$= \frac{1}{2}\|\omega\|^2 + \sum_{i=1}^{n}\alpha_i - \sum_{i=1}^{n}\alpha_i y_i \omega^T x_i - b\sum_{i=1}^{n}\alpha_i y_i$$

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b) KKT condition

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1) $y_i(\omega^T x_i + b) \geq 1 \quad i = 1, \dots, n$

2) $\alpha_i \geq 0 \quad i = 1, \dots, n$

3) $\nabla L \Rightarrow \nabla_\omega L = \omega - \sum_{i=1}^{n} \alpha_i y_i x_i = 0 \Rightarrow \omega = \sum \alpha_i y_i x_i$

$\nabla_b L = -\sum \alpha_i y_i = 0 \Rightarrow \sum \alpha_i y_i = 0$

4) $\alpha_i\left(1 - y_i(\omega^T x_i + b)\right) = 0 \quad i = 1, \dots, n \quad \begin{cases} \alpha_i = 0 \\ 1 - y_i(\omega^T x_i + b) = 0 \Rightarrow y_i(\omega^T x + b) = 1 \end{cases}$

Conditions:

1. $y_i(w^T x_i + b) \geq 1$
2. $\alpha_i \geq 0$
3. $w = \sum_{i=1}^{n} \alpha_i y_i x_i$, $\sum_{i=1}^{n} \alpha_i y_i = 0$
4. $\alpha_i(1 - y_i(w^T x_i + b)) = 0 \quad \forall i$

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c) Dual problem

$$\max \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum\sum \alpha_i \alpha_j y_i y_j (x_i^T x_j)$$

$$\text{s.t.} \sum_{i=1}^{n} \alpha_i y_i = 0 , \alpha_i \geq 0$$

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we have Lagrange: $L = \frac{1}{2} \|w\|^2 + \sum_{i=1}^{n} \alpha_i - \sum_{i=1}^{n} \alpha_i y_i w^T x_i - b \sum_{i=1}^{n} \alpha_i y_i$

dual function: $g(\alpha) = \inf_{w,b} L(w,b,\alpha) \rightarrow$ minimize over w and b for fixed α.

over b:

for term $b \sum \alpha_i y_i \rightarrow$ if $\sum \alpha_i y_i \neq 0$: by $b \rightarrow \pm\infty \rightarrow g(\alpha) = -\infty$

SO, $\sum \alpha_i y_i = 0$

over w: $\nabla_w L = w - \sum \alpha_i y_i x_i = 0 \rightarrow w^* = \sum \alpha_i y_i x_i$

$\Rightarrow \frac{1}{2} \|w^*\|^2 = \frac{1}{2} (w^*)^T (w) = \frac{1}{2} \left(\sum \alpha_i y_i x_i\right)^T \left(\sum \alpha_j y_j x_j\right) = \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j (x_i^T x_j)$

$\Rightarrow \sum_{i=1}^{n} \alpha_i y_i (w^*)^T x_i = \sum_{i=1}^{n} \alpha_i y_i \left(\sum \alpha_j y_j x_j^T x_i\right) = \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j (x_j^T x_i)$, $x_j^T x_i = x_i^T x_j$

$= \|w^*\|^2$

$\Rightarrow g(\alpha) = \sum_{i=1}^{n} \alpha_i + \frac{1}{2} \|w^*\|^2 - \|w^*\|^2 = \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \|w^*\|^2 = \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j (x_i^T x_j)$

$$\max \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j (x_i^T x_j)$$

$$\text{s.t} \sum_{i=1}^{n} \alpha_i y_i = 0 , \alpha_i \geq 0$$

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4 PCA

$X_1 = \{2.5, 0.5, 2.2\}$

$X_2 = \{2.4, 0.7, 2.9\}$

$X_3 = \{3.0, 1.5, 2.5\}$

$X_4 = \{1.0, 0.6, 1.1\}$

a) Covariance Matrix

$$\bar{x}_i = \frac{1}{N}\sum x_i^{(n)} \rightarrow \bar{x}_1 = (2.5+0.5+2.2)/3 = 1.7333,\ \bar{x}_2 = (2.4+0.7+2.9)/3 = 2.0000$$

$$\bar{x}_3 = (3.0+1.5+2.5)/3 = 2.3333,\ \bar{x}_4 = (1.0+0.6+1.1)/3 = 0.9000$$

$$x_c = x_i^{(n)} - \bar{x}_i \rightarrow X_1 = \{0.7667, -1.2333, 0.4667\},\ X_2 = \{0.4, -1.3, 0.9\}$$

$$X_3 = \{0.6667, -0.8333, 0.1667\},\ X_4 = \{0.1, -0.3, 0.2\}$$

$$S = \frac{1}{N}\sum x_n x_n^T = \frac{1}{3}\begin{bmatrix} 0.7667 & -1.2333 & 0.4667 \\ 0.4000 & -1.3000 & 0.9000 \\ 0.6777 & -0.8333 & 0.1667 \\ 0.1000 & -0.3000 & 0.2000 \end{bmatrix}\begin{bmatrix} 0.7667 & 0.4000 & 0.6667 & 0.1000 \\ -1.2333 & -1.3000 & -1.8333 & -0.3000 \\ 0.4667 & 0.9000 & 0.1667 & 0.2000 \end{bmatrix} =$$

$$S = \begin{bmatrix} 0.7756 & 0.7767 & 0.5389 & 0.1800 \\ 0.7767 & 0.8867 & 0.5000 & 0.2033 \\ 0.5389 & 0.5000 & 0.3889 & 0.1167 \\ 0.1800 & 0.2033 & 0.1167 & 0.0467 \end{bmatrix}$$

b) eigen value and eigen vector

$$Sb_m = \lambda_m b_m \rightarrow \det(S - \lambda I) = 0 \xrightarrow[\text{calculator}]{\text{use}} \lambda_1 = 1.9983,\ \lambda_2 = 0.0995,\ \lambda_3 = \lambda_4 = 0$$

$$\lambda_1 = 1.9983 \rightarrow b_1 = [-0.6183, -0.6479, -0.4191, -0.1496]^T$$

$$\lambda_2 = 0.0995 \rightarrow b_2 = [-0.3434, 0.6943, -0.6167, 0.1403]^T$$

$$\lambda_3 = 0 \rightarrow b_3 = [-0.2862, -0.1025, 0.2529, 0.9185]^T$$

$$\lambda_4 = 0 \rightarrow b_4 = [-0.6465, 0.2962, 0.6165, -0.3381]^T$$

c) Principal Component PC_1, PC_2

$$Z = B^T x \rightarrow PC_1 = b_1^T x_c,\quad PC_2 = b_2^T x_c$$

$$x_c^{(1)} = \begin{bmatrix} 0.7667 \\ 0.4000 \\ 0.6667 \\ 0.1000 \end{bmatrix} \rightarrow z^{(1)} = (PC_1^{(1)}, PC_2^{(1)}) = (b_1^T x_c^{(1)}, b_2^T x_c^{(1)})$$

$$= \begin{bmatrix} -0.6183 \\ -0.6479 \\ -0.4191 \\ -0.1496 \end{bmatrix}^T \begin{bmatrix} 0.7667 \\ 0.4000 \\ 0.6667 \\ 0.1000 \end{bmatrix}, \begin{bmatrix} -0.3434 \\ 0.6943 \\ -0.6167 \\ 0.1403 \end{bmatrix}^T \begin{bmatrix} 0.7667 \\ 0.4000 \\ 0.6667 \\ 0.1000 \end{bmatrix} = (-1.0275, -0.3827)$$

$$x_c^{(2)} = \begin{bmatrix} -1.2333 \\ -1.3000 \\ -0.8333 \\ -0.3000 \end{bmatrix} \rightarrow z^{(2)} = (PC_1^{(2)}, PC_2^{(2)}) = (b_1^T x_c^{(2)}, b_2^T x_c^{(2)})$$

$$= \begin{bmatrix} -0.683 \\ -0.6479 \\ -0.4191 \\ -0.1496 \end{bmatrix}^T \begin{bmatrix} -1.2333 \\ -1.3000 \\ -0.8333 \\ -0.3000 \end{bmatrix}, \begin{bmatrix} -0.3434 \\ 0.6943 \\ -0.6167 \\ 0.1403 \end{bmatrix}^T \begin{bmatrix} -1.2333 \\ -1.3000 \\ -0.8333 \\ -0.3000 \end{bmatrix} = (1.9989, -0.0072)$$

$$x_c^{(3)} = \begin{bmatrix} 0.4667 \\ 0.9000 \\ 0.1667 \\ 0.2000 \end{bmatrix} \rightarrow z^{(3)} = (PC_1^{(3)}, PC_2^{(3)}) = (b_1^T x_c^{(3)}, b_2^T x_c^{(3)})$$

$$= \begin{bmatrix} -0.6183 \\ -0.6479 \\ -0.4191 \\ -0.1496 \end{bmatrix}^T \begin{bmatrix} 0.4667 \\ 0.9000 \\ 0.1667 \\ 0.2000 \end{bmatrix}, \begin{bmatrix} -0.3434 \\ 0.6943 \\ -0.6167 \\ 0.1403 \end{bmatrix}^T \begin{bmatrix} 0.4667 \\ 0.9000 \\ 0.1667 \\ 0.2000 \end{bmatrix} = (-0.9714, 0.3899)$$

$Z_1 = -1.0275, 1.9989, -0.9714$

$Z_2 = -.3827, -.0072, 0.3899$

d) Singular values for Matrix X.

$$X = U\Sigma V^T \rightarrow XX^T = U(\Sigma\Sigma^T)U^T,\quad \Sigma\Sigma^T = \mathrm{diag}(\sigma_i^2) \rightarrow \lambda_i(XX^T) = \sigma_i^2$$

$$S = \frac{1}{N} XX^T \rightarrow \lambda_i(S) = 1/N\, \lambda_i(XX^T) \rightarrow \sigma_i^2 = N\lambda_i \rightarrow \sigma_i = \sqrt{N\lambda_i}$$

$$\sigma_1 = \sqrt{N\lambda_1} = \sqrt{3 \times 1.9983} = 2.4484,\quad \sigma_2 = \sqrt{N\lambda_2} = \sqrt{3 \times 0.0995} = 0.5464$$

$$\sigma_3 = \sigma_4 = 0$$

$\sigma_1 = 2.4484,\ \sigma_2 = 0.5464,\ \sigma_3 = \sigma_4 = 0$